Invest SA

South Australia

Building global partnerships in renewables and hydrogen



invest.sa.gov.au

SOUTH AUSTRALIA

Government of South Australia

Invest SA is a dedicated team of investment and business professionals within the Department of State Development, the only agency that can connect you with opportunities, industry networks, and other government agencies in the energy and mining sector in South Australia.

Invest SA, Minerals and Energy Team in collaboration with the Department for Energy and Mining offers project facilitation and case management services to develop your business case and to support your growth.

invest.sa.gov.au/sectors/energy-and-mining

The Department for Energy and Mining (DEM) leads the global transformation economy, overseeing the responsible mining and production of the minerals, metals and fuels of the future, to safely and sustainably generate the energy and low carbon products of the future.

energymining.sa.gov.au

South Australia's Energy Transition Partners

Australian Gas Infrastructure Group, agl, amp, Andromeda, Australian Rare Earths, Barton Gold, BHP

Boss Energy, Cobra Resources, Coda Minerals, CIP Copenhagen Infrastructure Partners, Digital Twinning Australia, elecseed, Energys

ENEOS, Enviro Copper, entX, epic energy, Fortescue, GFG Alliance, Green Gold Energy

Hallett Group, HDF Energy, HILT CRC, H2U, Hydrogen Park South Australia, Hyundai, Iberdrola, Investigator Resources Limited

Iron Road, iTech Minerals, ICHNP, Liberty, Linde, Ledestone Mines, Lot Fourteen

Magnetite Mines, Marubeni, MBH May Brothers Holdings, Mitsubishi Heavy Industries, MTE, Neoen

NeuRizer, Nexif Ratch, origin, osmofla, Port Augusta Operations, SPACE, RE global

Renascor Resources, res, REX Minerals Ltd, Rockburst Technologies, SA H2H Hydrogen Technology Cluster, Santos, SAPGEN Summerfield Renewable Energy Hub, Scaling Green Hydrogen CRC

Siemens Energy, Tesla, The University of Adelaide, tlf renewables, tindo solar, Trafigura, Vestas, zero petroleum

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Our strengths = your opportunity



AUD$21BN pipeline of renewable energy projects



World class renewable and mineral resources



Integrated clean energy network



Technology and commodity agnostic



A world leading regulatory system



Manufacturing opportunities



Cooperative research centres on industry decarbonisation



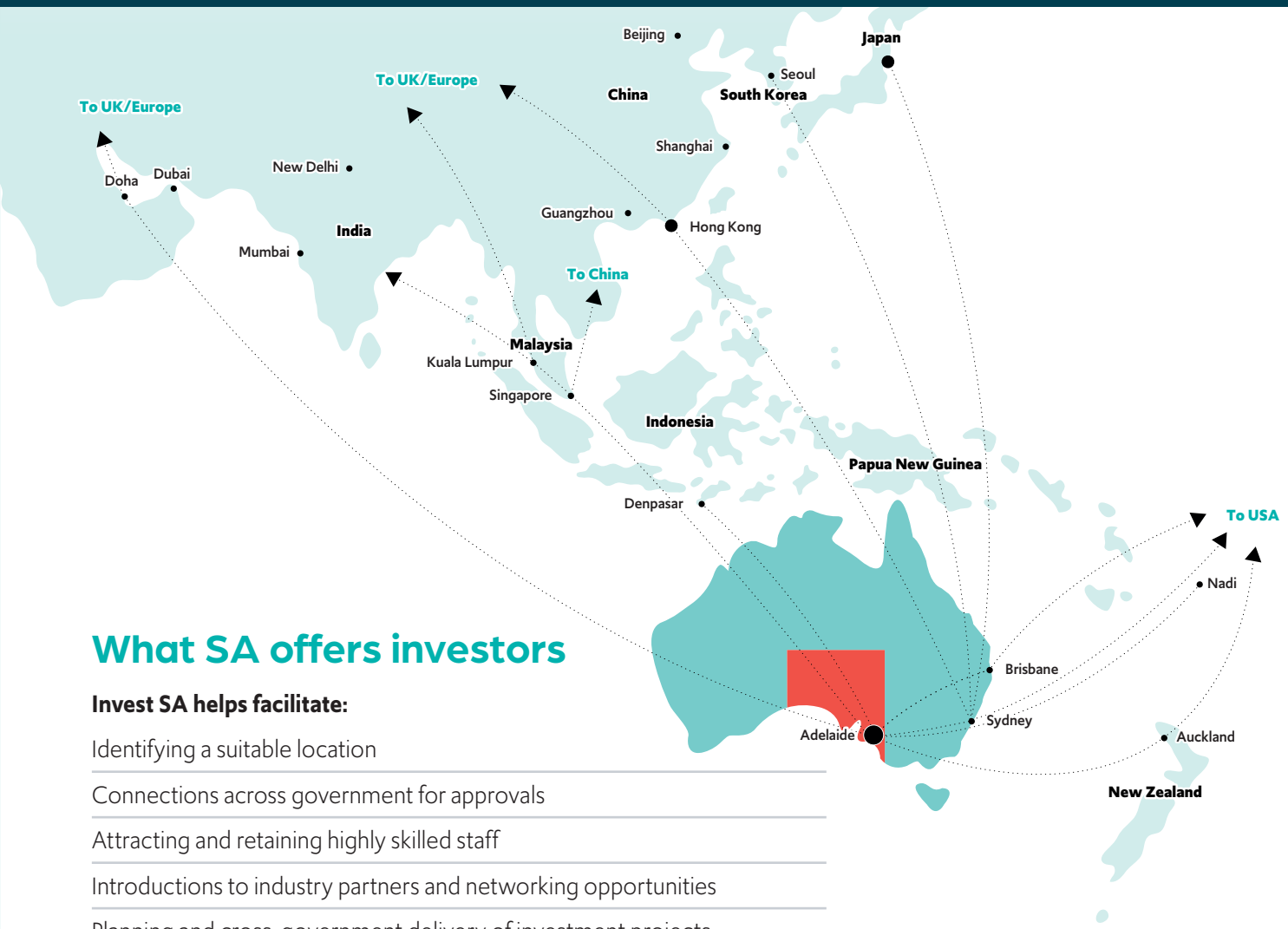
Collaborate with local stakeholders to build your supply chain

"For close to 10 years, Neoen has been a major investor and operator of renewable energy generation and storage in South Australia – with more than 1 gigawatt of capacity now under construction or operating. Our desire to continue to invest in the state is driven not only by the availability of world-class renewable resources, but also through the maturity of the local industry and the ability to partner effectively with local and state government agencies."

Louis de Sambucy, Managing Director Neoen Australia

Introducing South Australia

Competitive construction costs	1.8 million people in South Australia
Competitive low tax business environment	+9.30hr difference from UTC
Supportive investor environment	Strong community support
Australian mainland leader in staff retention	Committed to 500% renewables by 2050
Total land area of 983,482 square kilometres	Premium city office accommodation
Shares borders with all mainland states with existing infrastructure corridors	A more liveable city



What SA offers investors

Invest SA helps facilitate:

- Identifying a suitable location
- Connections across government for approvals
- Attracting and retaining highly skilled staff
- Introductions to industry partners and networking opportunities
- Planning and cross-government delivery of investment projects
- Connections with our global teams and identifying export opportunities

Invest with Australia's renewable energy, hydrogen, and minerals capital

Hydrogen unlocks value

South Australia is an international leader in integrating wind, solar, and battery energy system into the state's energy network. In 2023, over 74% of South Australia's electricity generation came from wind (47%) and solar (27%).

Abundant wind and solar resources, land availability, existing and developing infrastructure, and skills combine to place South Australia in prime position to be a world-class producer, user, and exporter of hydrogen.

It is estimated there are over 100 gigawatts (GW) of wind and solar resources in South Australia with further development potential – that's more than 10 times the state's current registered electricity generation capacity.

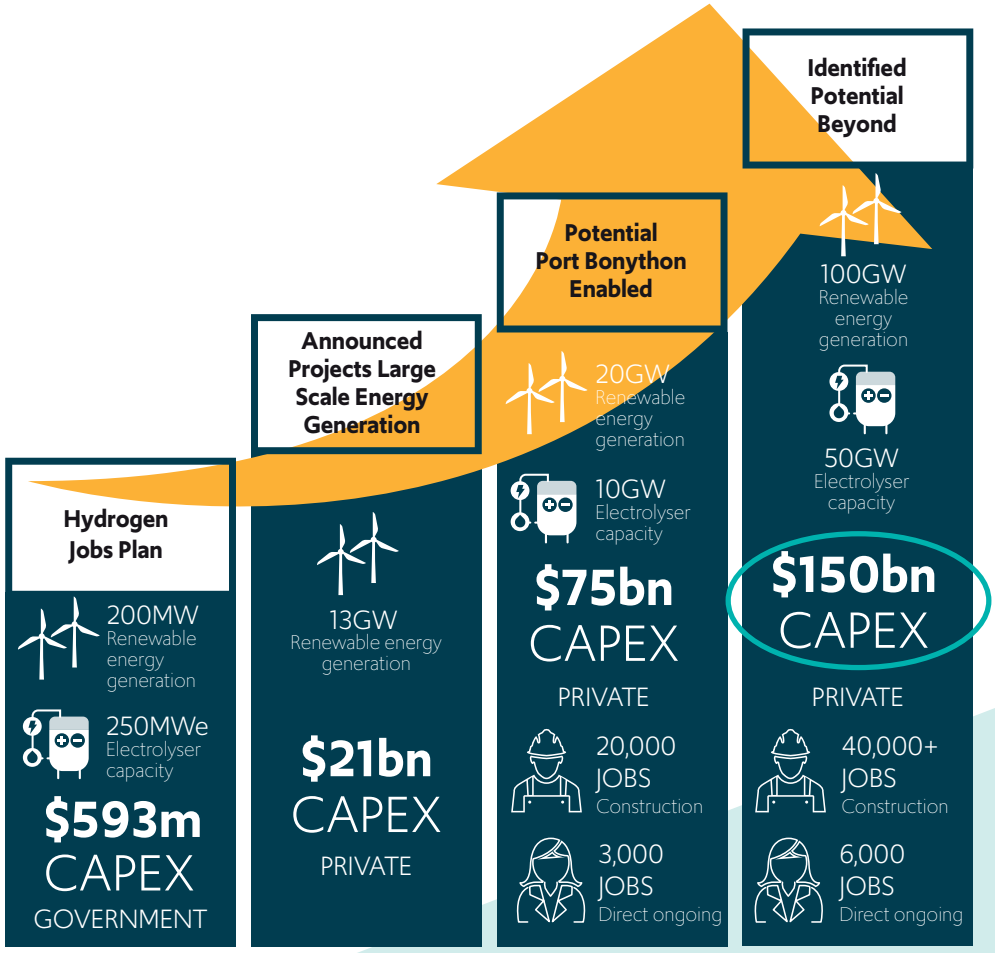
South Australia is on track to meet 100% of our electricity demand from renewable energy by 2027.

Our low-cost power, reputation for innovation, and world-leading regulatory system is leading the decarbonisation of industry and modern manufacturing opportunities.

"South Australia's renewable energy resources initially attracted global energy transition platform Amp Energy to South Australia. The state's clear commitment to leading the Green Hydrogen Economy and the continuous support from the South Australian Government ultimately led Amp to establishing the Renewable Energy Hub, together with Amp's green hydrogen and ammonia production and storage expansion, in South Australia with a total expected investment in excess of AUD$2 billion.

We look forward to continuing this productive relationship with the South Australian Government and to realise Amp's green hydrogen and ammonia ambitions."

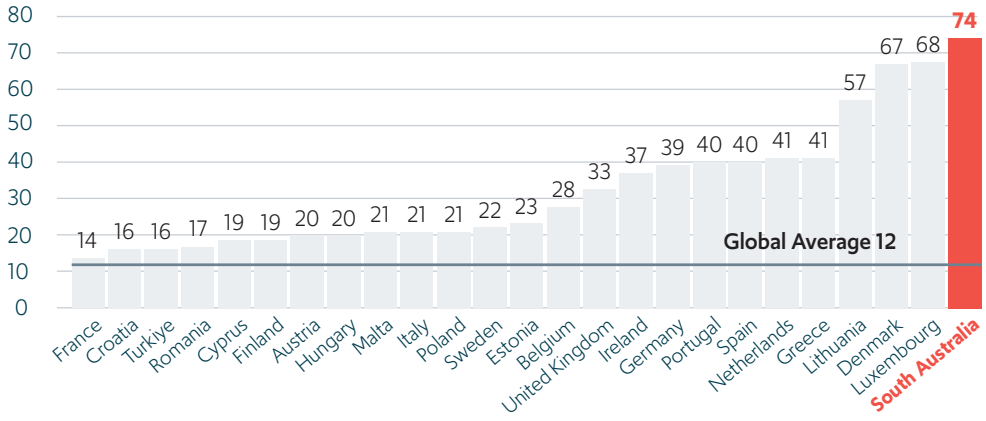
Paul Ezekiel, Co-Founder and CIO, Amp Energy



South Australia ranked **number one globally** for annual renewable energy



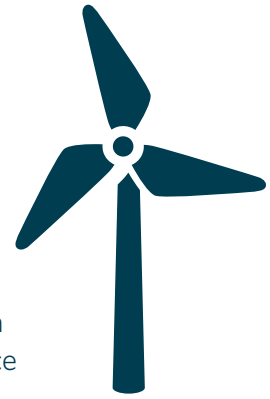
% share of wind and solar in electricity generation by market, 2023



Source: opennem.org.au

Projects in operation

- 24 wind farms
- 5 solar farms
- 6 grid-scale batteries
- Australia's largest hydrogen electrolyser (operating since May 2021)
- South Australia is leading the world with 45% of homes having rooftop solar

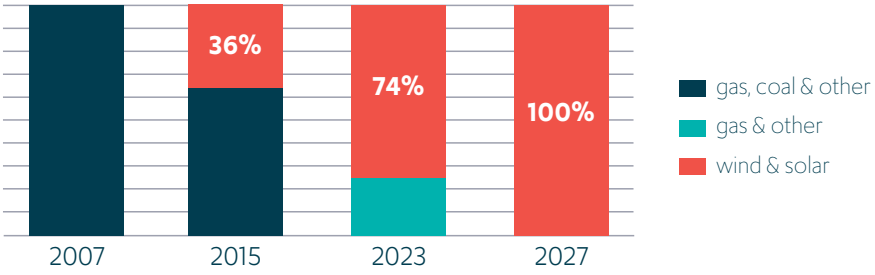


56 **Renewable energy projects**

in the development pipeline totalling

$21bn **investment**

On track for net 100% renewables by 2027

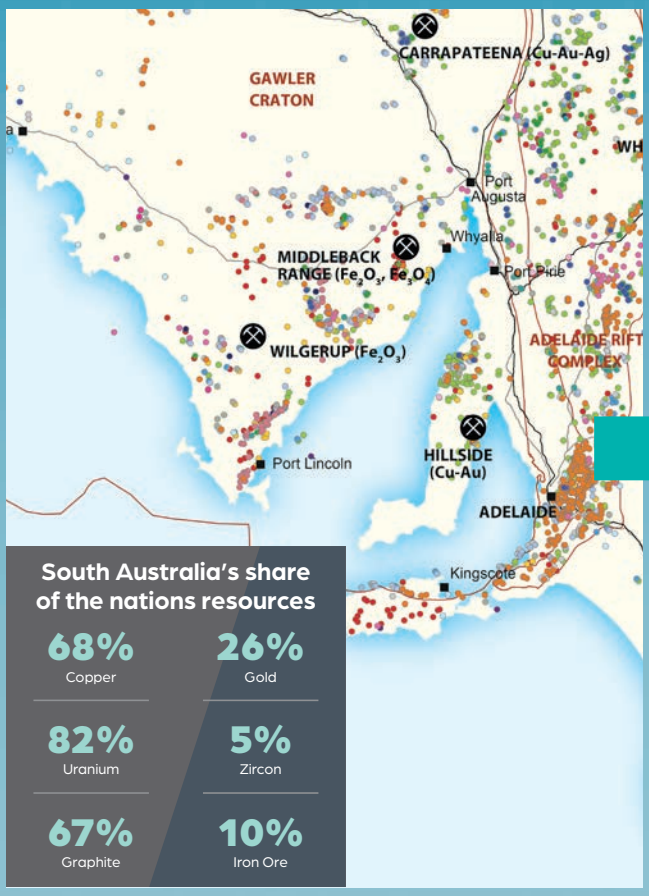


"South Australia is leading the energy transition in Australia. This demonstrates how the diligent application of innovative technologies can fulfil global sustainability objectives, while still being safe, reliable, and cost effective. Most importantly, this increases customer confidence and helps pave the way for the increased uptake and implementation, thus accelerating the speed of the energy transition."

Samuel Morillon, Managing Director of Siemens Energy in Australia

South Australia is set for a decarbonised future

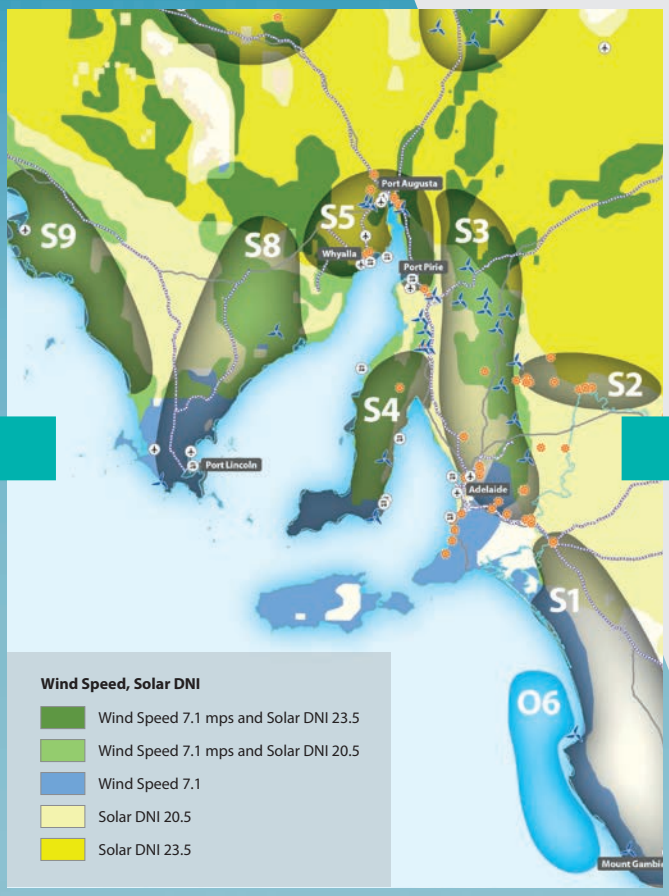
Mineral resources



South Australia's share of the nations resources

68% Copper	**26%** Gold
82% Uranium	**5%** Zircon
67% Graphite	**10%** Iron Ore

Renewable Energy Zones (REZ)





Port Augusta Renewable Energy Park

Infrastructure corridors



Energy hubs



Adelaide (HyP SA, Tonsley)


Port Bonython (Port Bonython Hydrogen Hub)


Whyalla (Hydrogen Jobs Plan, Green Iron / Steel Manufacturing)


Cape Hardy Industrial Precinct (Desalination Plant, Hydrogen Production Facility and potential for green minerals)


Port Augusta (Green Cement Facility)

Developing South Australia's net-zero carbon future

1. On track to reach **100% renewable** generation by 2027.
2. Rising force for **green minerals** (copper, magnetite, graphite, cobalt, zircon and rare earth elements).
3. Renewable **hydrogen superpower** by delivering the 'Hydrogen Jobs Plan' (250MWe electrolyser + 200MW power plant + storage facility) and supporting hydrogen hub developments at Port Bonython and Cape Hardy.
4. The Northern Water project will unlock growth in mining, defence, hydrogen and pastoral industries, in regions **critical to achieving net-zero** targets, through construction of a seawater **desalination plant** that connects up to a 600 kilometre transfer pipeline from the Spencer Gulf to northern South Australia.

Decarbonisation requires renewables, which requires resources!

South Australia is a global leader in strategic and critical mineral exports with Australia's richest copper resource and the world's largest zircon mine, Jacinth-Ambrosia. Our state is also home to the world's second largest proven graphite reserve and promising cobalt resources, which are crucial for lithium iron battery production, and has an expanding inventory of rare earth elements used in high temperature magnets.

Demand for minerals, such as copper, cobalt, and graphite, will grow quickly, as the global energy transition gathers pace. This demand will create significant opportunities for South Australia due to our critical mineral reserves and our reputation as a trusted and reliable supplier.

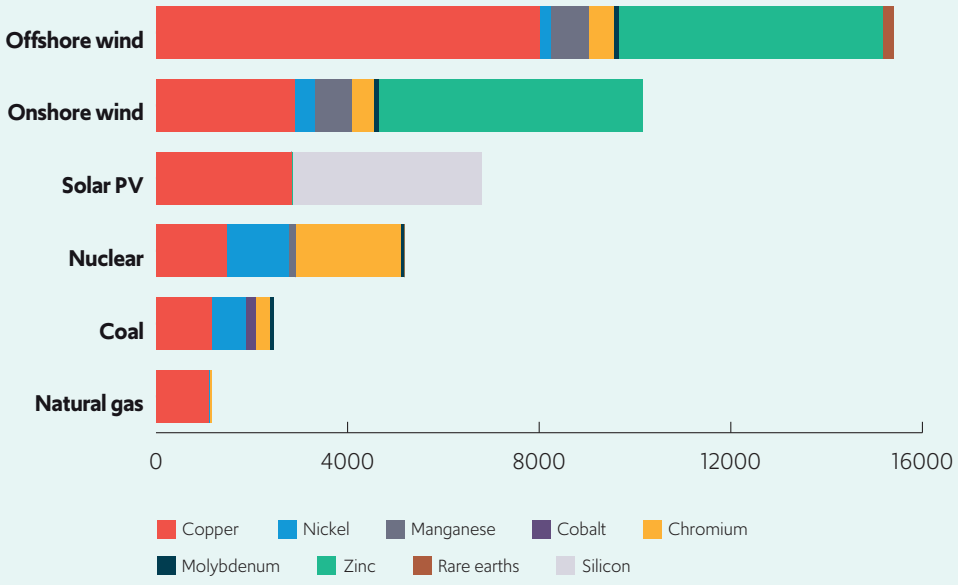
South Australia has huge potential to play an important role in industry supply chains with the superior quality magnetite to assist the global steel industry to decarbonise, and high-purity alumina for a range of technologies, and the manganese and silicon resources as the main alloying elements used in carbon steel production.

The Government of South Australia supports the Australian Government's visions for strengthening Australia's sovereign capabilities and increasing the capacity of downstream manufacturing. We are keen to contribute through our streamlined regulatory framework and capabilities, while adding value domestically, rather than exporting the high-value raw materials as unprocessed commodities.

The South Australian Drill Core Reference Library, located in the Tonsley Innovation District, hosts geological samples recovered from over 130 years of exploration for minerals and energy resources otherwise hidden beneath the land surface of South Australia. These irreplaceable samples represent valuable direct records of the geological materials retrieved from the depths of the crust, that will continue to be investigated and analysed by new generations of geologists in industry, research, education and government.

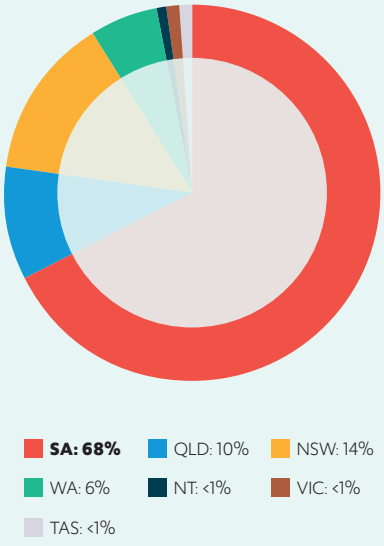
The Government of South Australia recognises the link between mining and renewable energy. This link is critical to the world's net-zero emissions future.

Minerals used in clean energy technologies compared to other power generation sources



Source: IEA (2021), The Role of Critical Minerals in Clean Energy Transitions

South Australia contains 68% of Australia's economic demonstrated resources of copper



Source: Geoscience Australia

State Prosperity PROJECT

We have what the world wants

The global energy transition will drive the world's economy for decades to come, and with our unique combination of sun, wind and valuable minerals in the Upper Spencer Gulf and surrounds, we have what the world wants. What the world demands.

South Australia's renewable energy share has surged from 1% in 2007 to 74% in 2023. We are Australia's renewable energy superpower.

South Australia is also home to Australia's richest copper resource, with global demand for copper set to surge for use in sustainable technology such as electric vehicles and wind turbines.

The Upper Spencer Gulf has superior magnetite iron ore resources which is essential for the production of green steel, with demand set to surge as the world seeks to decarbonise.

The key to unlocking the full potential of our critical minerals is water, which is why the State Government and private sector are progressing plans for a desalination plant and pipeline network called Northern Water.

H_2 The State Government is seeking to harness our renewable energy resources into hydrogen, by building the world's largest Hydrogen Power Station at Whyalla. Hydrogen is also essential to the production of green steel.

Collectively, these measures have the potential to generate billions of dollars of economic activity and create thousands of jobs.







South Australia's Green Hydrogen Electrolyser and Power Station

The state government has committed funding of more than half a billion dollars to build a world-first grid-scale green hydrogen facility by 2025. The facility will include 250 megawatts of electrolyser, 200 megawatts of power generation, and hydrogen storage infrastructure.

This investment will accelerate the growth of South Australia's hydrogen economy and help to unlock a gigawatt scale pipeline of further renewable energy developments across the state.

The Government of South Australia is also leading a consortia of industry partners committed to the development of South Australia's first large-scale hydrogen export terminal.

Port Bonython Hydrogen hub – primed for export

This South Australian Government-led Port Bonython export hub in the Upper Spencer Gulf will create a large-scale clean hydrogen production precinct for both export and domestic markets.

The Port Bonython Hub will create an economy of scale to drive down the cost of production, unlocking further demand for hydrogen as prices fall. The proposed projects at Port Bonython currently represent a total investment of about AU$13 billion and have potential to generate up to 1.8 million tonnes of hydrogen.

Cape Hardy

Cape Hardy is located on the east coast of Eyre Peninsula, South Australia, approximately 7 kilometres south of Port Neill. It comprises 1,200 hectares of undeveloped coastal land, with naturally occurring deep-water, wholly owned by Iron Road Limited (ASX:IRD).

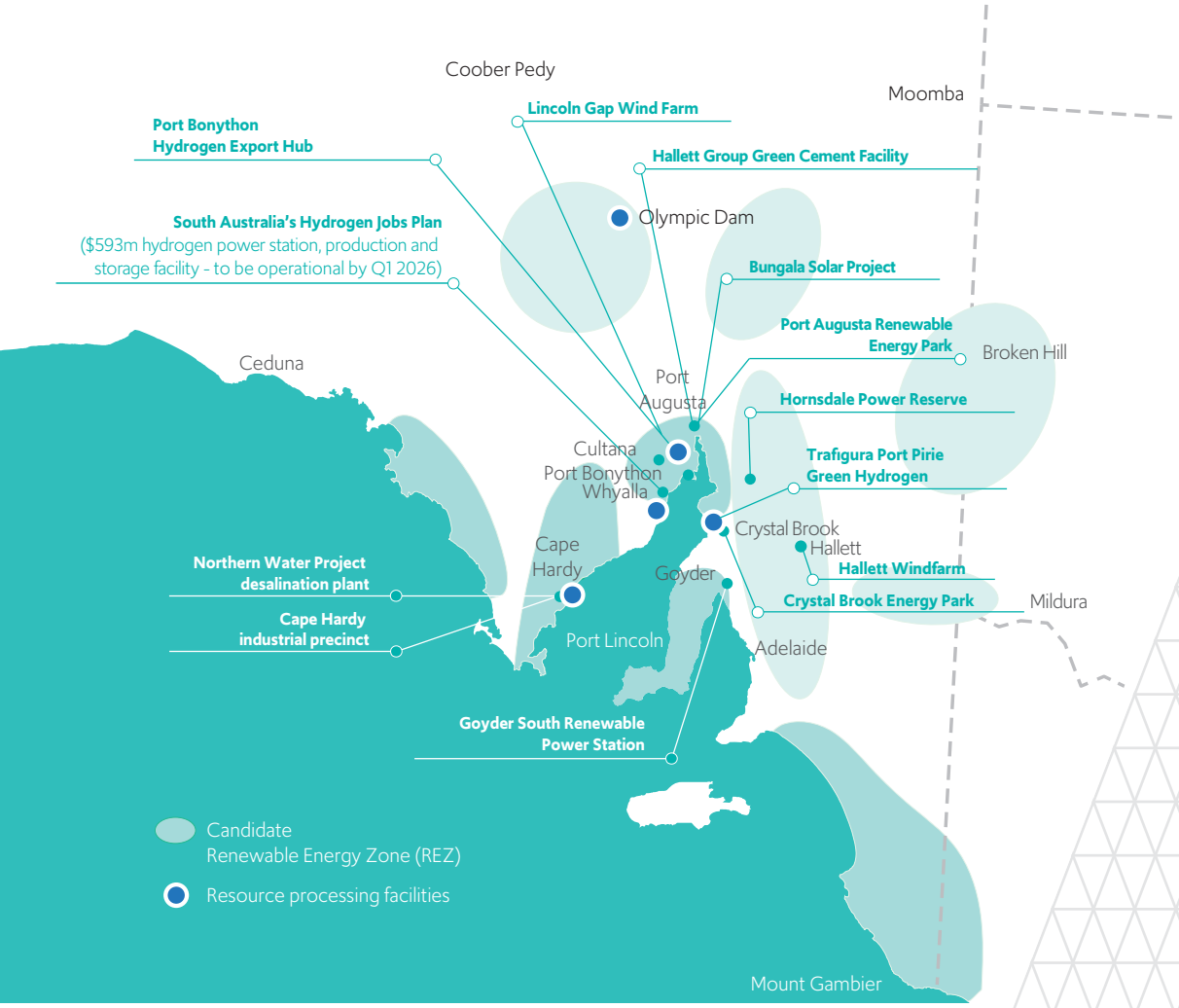
Cape Hardy is considered to have excellent potential as a green hydrogen development hub in South Australia and will include an industrial, cape-size port precinct with multi-commodity capability (including hydrogen, grain, and minerals).

Northern Water Project

Aiming to provide a new, climate independent water source for the Far North, Upper Spencer Gulf and Eastern Eyre Peninsula regions of South Australia, the Northern Water Project would enable the growth of industries crucial to achieving net-zero goals, including the emerging green energy and hydrogen industries.

This project would involve construction and operation of a 260ML/day seawater desalination plant (in two stages) in the Spencer Gulf connected to northern South Australia via up to 600 kilometres of pipeline.

The final decision to proceed with construction of the Northern Water project will rely on gaining required project approvals and agreements. The final project decision is expected in 2025.



A global innovator



Photo credit: Neoen Australia

South Australia hosts the world's original big battery at Neoen's Hornsdale Power Reserve.

South Australia is home to Australia's largest operating electrolyser at Hydrogen Park SA that has been blending renewable hydrogen into the local gas network, and is supplying renewable hydrogen for industrial use in the Upper Spencer Gulf area.



Hydrogen Park SA, AGIG

World leading regulation

South Australia is an advocate for internationally harmonised codes and the highest standards to ensure beneficial safety and environmental outcomes from hydrogen and renewable energy developments.

It is a member of the International Association for Hydrogen Safety, HySafe, with committee representation at the highest level.

South Australia is among the first regions to prepare a dedicated regulation to provide certainty for investors while building community confidence.

The Hydrogen and Renewable Energy Act 2023 passed the South Australian Parliament in November 2023, which is the world's first legislative framework designed to provide a coordinated approach to the burgeoning hydrogen and renewable energy industries. The Act streamlines the process for companies wanting to invest in large-scale hydrogen and renewable energy projects in South Australia, bringing issues such as land access, environmental impacts, and native title rights into a single regulatory process. The Act ensures extensive community and landholder consultation is carried out, so that hydrogen and renewable energy projects meets communities' expectations of responsible development.

Leading research capability

As a frontier location, the state has the best minds and scientific organisations collaborating to advance hydrogen and renewable technologies and bring costs down.

The government and industry have forged centres of excellence and internationally significant partners to transition industries to reduce carbon emission.

South Australia welcomes the opportunity hydrogen production brings to become the world's proving ground to transition energy-intensive mineral processing towards carbon neutrality on a competitive level.

Trusted international trade and supply chain relationships

South Australia is a dynamic partner, actively engaging with international hydrogen consortia to share information and cooperate on the scale-up of the global renewable and hydrogen production export industry.

With well-established trade partnerships, the state is well positioned to supply markets across Asia and Europe, with existing trade relationships.

"With a history of leading the development of renewable energy technology, South Australia offers a supportive investment environment, strong wind speeds, high quality solar and great renewable development conditions to build the hydrogen industry."

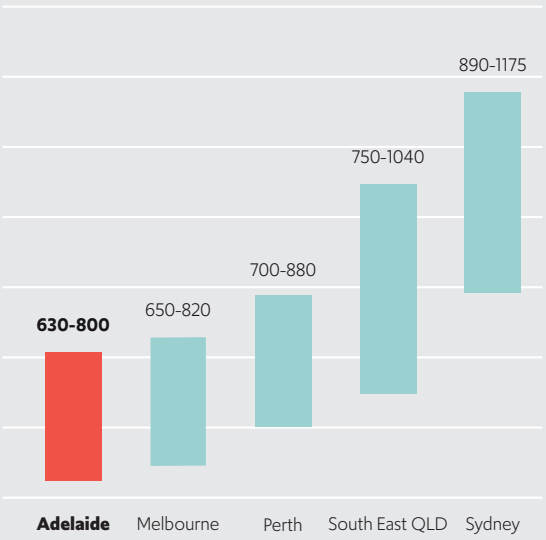
Origin Energy, Export and Infrastructure

Australia's most cost competitive city

South Australia's competitive business environment supports commercial success and scaling of operations.

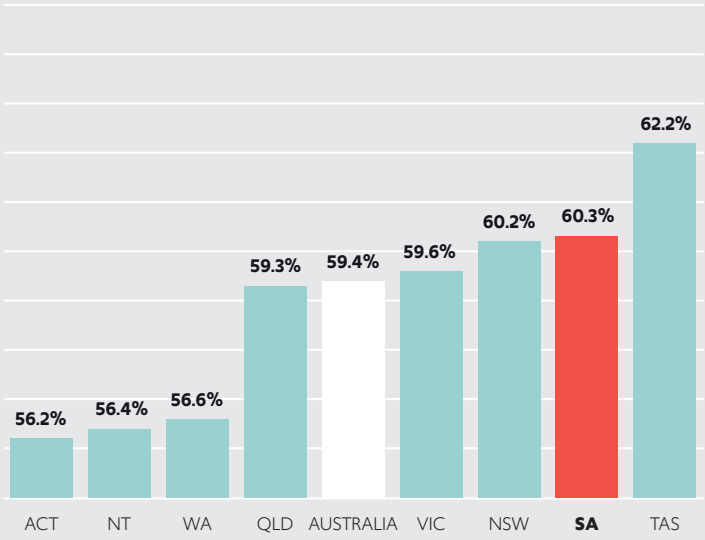
South Australia offers a suitably skilled and scalable workforce with competitive construction costs and high staff retention rates.

Industrial construction costs (AUD/sqm)



Source: Napier & Blakely Construction Costs Datacard, July 2022
**Cost estimated for Up to 10m high warehouse - medium standard, precast walls, sprinklers (over 3,000 sqm)*

Staff retention rates



Source: Australian Bureau of Statistics - Job Mobility [Table 12 - Labour mobility, retrenchments and duration of employment by state and territory, February 2022]



Innovation districts

Our sectors co-exist in hi-tech hubs and research centres, forging collaborative ecosystems that open doors and inspire new ways of thinking.

Defence Super District
34 kms from CBD

Key national defence research, manufacturing and sustainment hub housing the RAAF Base Edinburgh, Defence Science and Technology Group and major defence companies.

Osborne Naval Precinct
26 kms from CBD

Australia's largest naval shipbuilding hub incorporating a critical mass of world-class warship design and construction skills. Significant upgrades are slated to make it the most technologically advanced naval shipyard in the world.

Technology Park
17 kms from CBD

Hub for systems development and integration, information communications technology, advanced manufacturing and electronics, and home to over 100 small to medium enterprises, multinationals and start-ups.

Adelaide BioMed City
Adelaide CBD

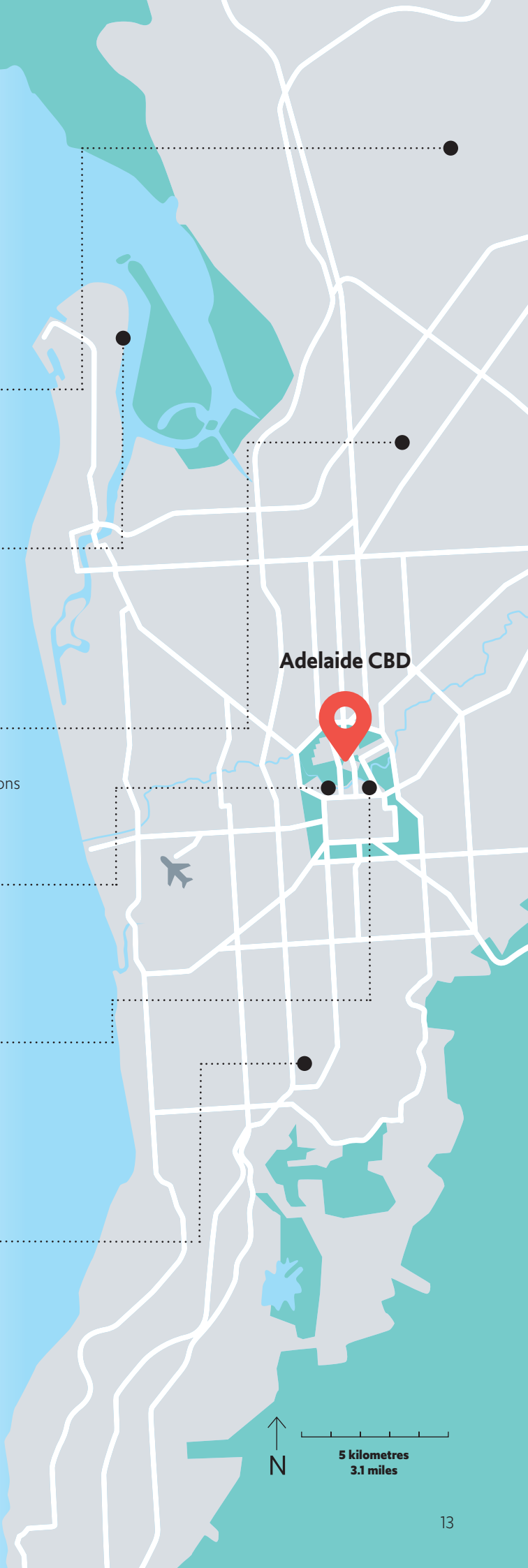
AUD$3.8 billion health and medical innovation district with leading edge anchor institutes and companies that cluster and connect with start-ups, business incubators and accelerators.

Lot Fourteen Innovation District
Adelaide CBD

Australia's first ideas and innovation neighbourhood and home to the Australian Space Agency, SmartSat CRC, Defence and Space Landing Pad, FIXE start-up hub, Adelaide, MIT Adelaide Living Lab, Australian Institute for Machine Learning and the Australian Cyber Collaboration Centre.

Tonsley Innovation District
12kms from CBD

Home to high-value manufacturing industries including defence, health, medical devices and assistive technologies, clean tech and renewable energy, software and simulation, mining and energy services.



Advantage Adelaide – live, work & play

South Australia is a great place to live, work and play. The capital Adelaide is consistently ranked as one of the most liveable cities in the world.

Adelaide is a progressive, cosmopolitan capital offering a premium lifestyle with world renowned festivals, major sporting events and iconic locations, such as the Barossa Valley, Kangaroo Island and Flinders Ranges.

South Australia's innovative, cost competitive and agile nature is luring more young talent, and our ability to rapidly scale and provide skilled workers in critical sectors has caught the attention of world-class companies.

We're forging global partnerships and leveraging our position as one of the most liveable and safest cities to grow our state and opportunities for everyone.

There's a strong commitment to collaboration among industries with specialist precincts, such as the Tonsley Innovation District and Lot Fourteen, encouraging companies to thrive.

- **Shortest daily work commute of the major mainland cities – 21% less than Sydney [1]**
- **One of the lowest cost of living capital cities in Australia [2]**
- **Adelaide's median house price is $680,000 – 44% lower than Sydney[3]**

[1] *Melbourne Institute: Applied Economic & Social Research, The Household, Income and Labour Dynamics in Australia Survey*

[2] *Mercer 2023 Cost of Living city ranking – Sydney 58, Melbourne 67, Brisbane 84, Perth 97, Adelaide 102*

[3] *Australian Bureau of Statistics, March quarter 2023*



Riverbank Precinct, Adelaide
Image courtesy of Olivia Dowling



Adelaide Central Markets
Image courtesy of South Australian Tourism Commission

The South Australian lifestyle

There is a year-round calendar of large events, entertainment, culture and excitement. Some of the best known cultural and arts events include the Adelaide Festival, WOMADelaide, Tasting Australia, Adelaide Fringe, Adelaide Cabaret Festival and OzAsia Festival. South Australia hosts international and national sporting events including cycling's Tour Down Under and the Adelaide 500 car race.

Our Mediterranean climate – hot, dry summers and mild, wet winters – enables an exciting outdoor lifestyle. South Australia also hosts globally-renowned sporting events including LIV Golf Adelaide, winner of the World's Best Golf Event of the Year 2023 at the world golf awards.

Adelaide's coastline stretches more than 30 kilometres, delivering wide, golden beaches with clean seas for swimming, kayaking, surfing and more.

It's hard not to escape into nature with parks and walking trails everywhere.

Best of all, getting around Adelaide is effortless. Nothing is really more than 30 minutes away by car so it's easy to have a family, sporting and social life every evening and an action packed, or relaxing, weekend!

South Australia delivers a better, more affordable lifestyle and opportunities – whether in the city or suburbs, on the coast or in the hills.

- **Most affordable unit rents and third most affordable house rents of any Australian capital city[4]**
- **Affordable and accessible childcare**
- **Good public transport and no toll roads**
- **Great public and private schools**
- **Adelaide's three public universities are ranked in the top 1% of the estimated 26,000 higher education institutions in the world.[5]**

[4] *Domain Rental Report, June 2023*

[5] *Source: Times Higher Education World University Rankings 2024 and qs.com*





Garden of Unearthly Delights
Image courtesy of South Australian Tourism Commission





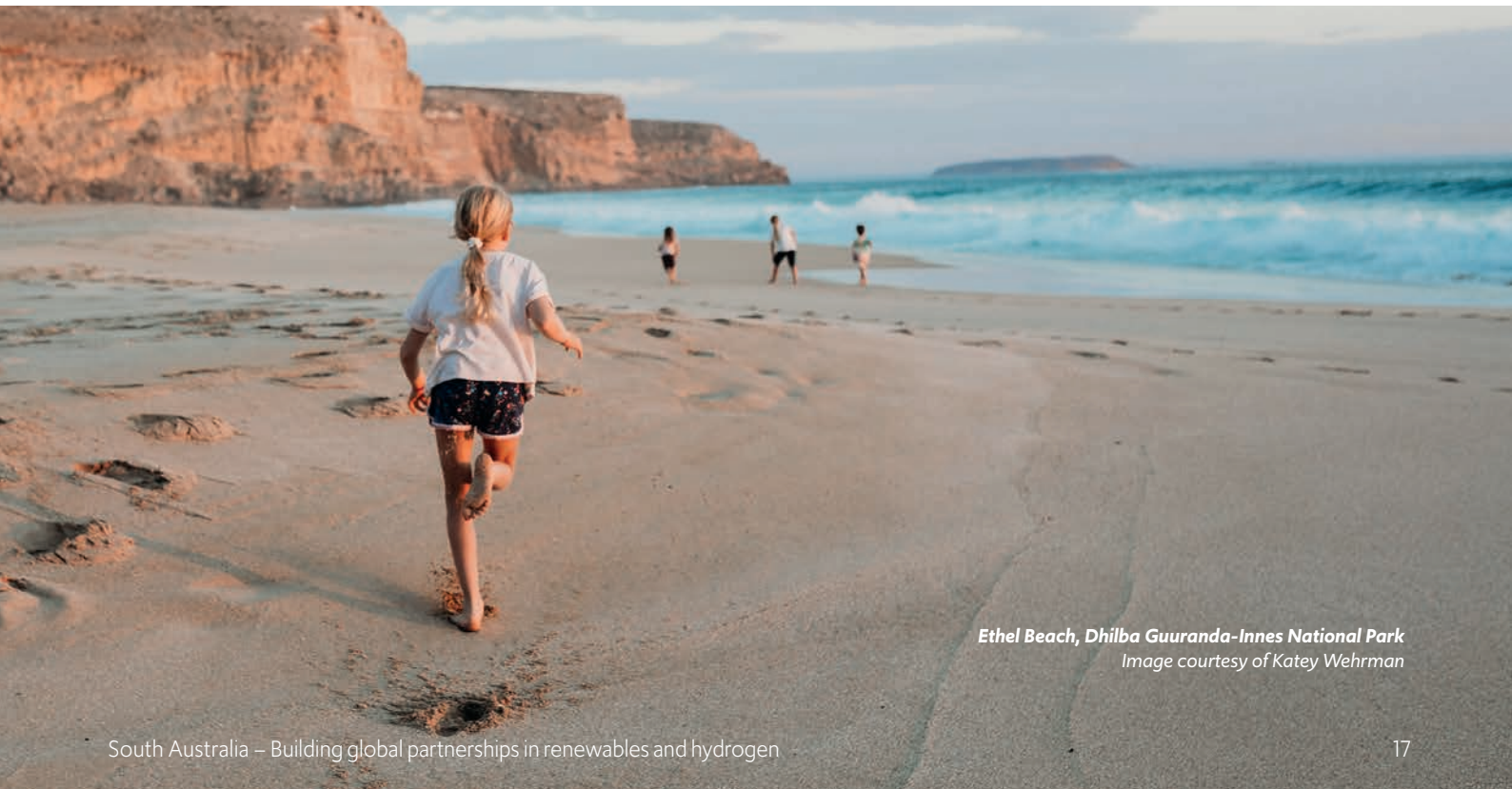
d'Arenberg Cube, McLaren Vale
Image courtesy of South Australian Tourism Commission/ d'Arenberg Cube



WOMADelaide
Image courtesy of Grant Hancock



LIV Golf Adelaide
Image courtesy of Chris Trotman/LIV Golf



Ethel Beach, Dhilba Guuranda-Innes National Park
Image courtesy of Katey Wehrman

Let's talk

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